

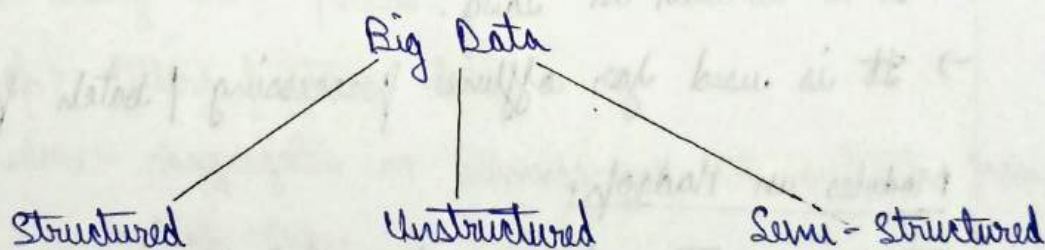
1) What is Big data? Explain types of Big Data with example.

Ans → Data Qualities, Characters, symbols on which operations are performed by computer which may be stored & transmitted in form of electrical signals & record on magnetic optical recording media like CD is called data.

→ Big Data It is also data with huge size growing exponentially with time.

It is a collection of large datasets that can't be processed using traditional tools efficiently.

There are three types of Big Data:-



(i) Structured Big Data

A data that can be stored & processed in a fixed format

Ex Employee table in database.

(ii) Unstructured Big Data

A data that can be stored without any fixed format.

Ex Google Search.

(iii) Semi-structured Big Data

A data with combination of both structured & unstructured big data

Ex XML (Extensible Markup Language).

2) What is Hadoop? Explain various models of Hadoop.

Ans → Hadoop is an open source from Apache.

→ It is used to store, process and analyze the data which is in very large volume.

→ It is written in Java.

→ It is used for offline processing / batch processing.

Modules in Hadoop

There are two types of modules in Hadoop:-

(i) HDFS (Hadoop Distribution File System)

(ii) Map Reduce.

(i) HDFS

A distributed file system that stores data across multiple machines.

→ Splits large files into blocks & distributes them across clusters.

→ Each block is replicated for fault tolerance.

→ It has two major components:-

① Name node:- Stores metadata

② Data node:- Stores actual data blocks.

(ii) Map Reduce

It is the data processing engine of ~~the~~ Hadoop.

→ It processes large datasets in parallel & distributed manner using two phases:-

① Map:- Filters & stores data

② Reduce:- Aggregates or summarizes the output from Map phase.

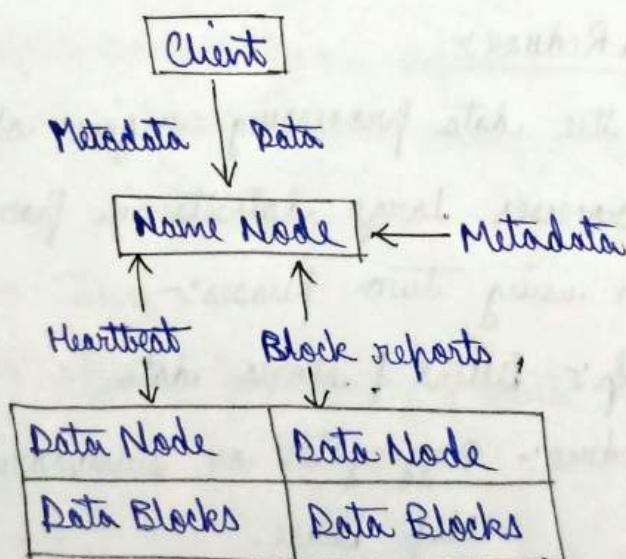
→ It runs on top of yarn for resource management.

3) What is HDFS? Explain components of HDFS.

Ans → Hadoop Distributed File System is a primary storage of Hadoop designed to store & manage large files across distributed cluster of machines.

→ It follows a master-slave architecture & provides high fault tolerance, high throughput & scalability making it ideal for Big Data applications.

Architecture of HDFS



Main Components

(i) Name Nodes

→ Brain of HDFS

→ Manages metadata of file system:

- File names
- Directories
- Block locations
- Access permissions

→ Does not store actual data, only metadata.

(ii) Data Node (Slave node) :-

- Stores actual node blocks on local disk
- Periodically sends heartbeat & block reports to Name Node
- Responsible for serving read/write requests from clients

(iii) Client :-

- Interface for users & applications to interact with HDFS.
- Communicates with Name Node to get block metadata.
- Connects directly with data nodes to read & write data.